Cambridge Assessment International Education

Cambridge O Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

9388981624

MATHEMATICS (SYLLABUS D) **4024/11**

Paper 1 **May/June 2021**

2 hours

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You may use tracing paper.
- You must show all necessary working clearly.

INFORMATION

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages.

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[Turn over

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER

1 Work out.

(a) $74.6 \times 10 - 3.89 \times 100$

..... [1]

(b) $5 + 3 \times 2 - 1$

..... [1]

2 15 125 $\sqrt{8}$ 11 $\sqrt{25}$ 14 60

From the numbers above, write down

(a) a factor of 70,

..... [1]

(b) a cube number,

..... [1]

(c) an irrational number.

..... [1]

3 **(a)** Work out $\frac{3}{7} + \frac{2}{5}$.

..... [1]

(b) Find $\frac{2}{3}$ of $\frac{6}{11}$, giving your answer as a fraction in its simplest form.

..... [1]

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4 **(a)** A record is kept of the water level in a harbour.
One morning, the level is 5 m. That afternoon, the level is −2 m.

Find the difference between the level in the morning and the level in the afternoon.

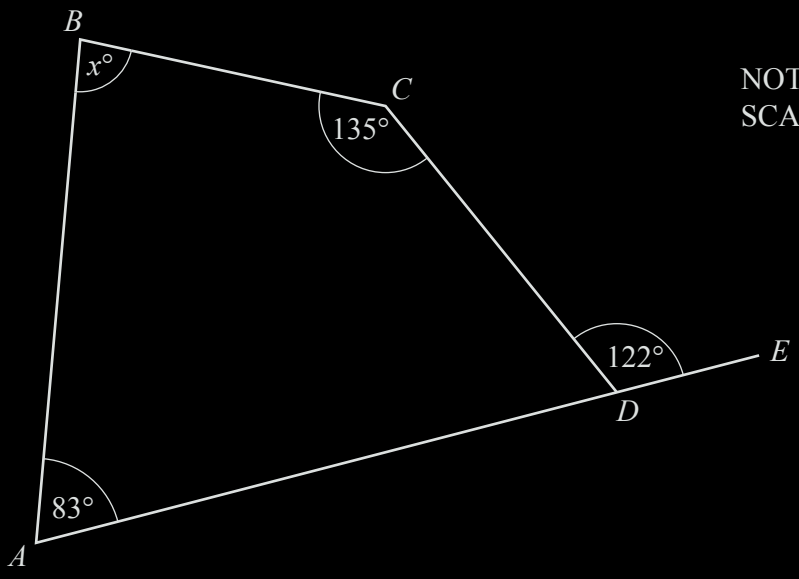
..... m [1]

(b) One day, the temperature at midday is 9 °C.
At midnight the temperature has dropped by 15.3 °C.

Find the temperature at midnight.

..... °C [1]

5



The diagram shows quadrilateral $ABCD$ with AD extended to E.
Angle $BCD = 135°$, angle $BAD = 83°$ and angle $CDE = 122°$.

Find the value of x.

$x =$ ..... [2]

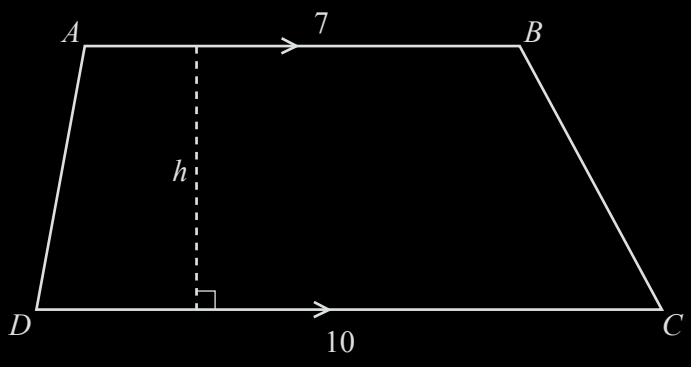
6 **(a)** Write 308 as a product of its prime factors.

..... [2]

(b) Find the highest common factor (HCF) of 308 and 66.

..... [1]

7



NOT TO SCALE

The diagram shows trapezium $ABCD$.
$AB = 7$ cm and $DC = 10$ cm.
The area of $ABCD$ is $85\,\text{cm}^2$.
The perpendicular height of the trapezium is h cm.

Find the value of h.

$h =$ ..... [2]

8 **(a)** Simplify $6x + 15 - 2x + 8$.

..... [1]

(b) Expand and simplify $(x - 5)^2$.

..... [2]

9 Insert the correct symbol $=$, $>$ or $<$ to make each statement correct.

(a) 0.6 kg ..... 60 g [1]

(b) 15 km ..... 15 000 m [1]

(c) $4\,\text{m}^2$ ..... $400\,\text{cm}^2$ [1]

10 By writing each number correct to one significant figure, estimate the value of

$$\frac{362.4 - 187.2}{52.3}.$$

..... [2]

11 **(a)** In a survey, 3 out of every 100 women were taller than 1.9 m.
One of these 100 women is picked at random.

Calculate the probability that she is **not** taller than 1.9 m.

..... [1]

(b) A new housing estate is being planned.
There are three possible plans: *A*, *B* and *C*.
A survey was carried out to see which plan people preferred.
The relative frequency table shows the results.

Plan	*A*	*B*	*C*
Relative frequency	0.3	0.5	0.2

52 people preferred plan *C*.

(i) Find how many people preferred plan *A*.

..... [2]

(ii) Calculate the total number of people surveyed.

..... [1]

12 Bernard bought a game in the USA for $15.
Alice bought the same game in Zambia and paid the equivalent price in Zambian kwacha (ZK).

Exchange Rate

1ZK = $0.075

Calculate the price that Alice paid.

..... ZK [2]

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13 Two numbers x and y are such that

- $x : y = 5 : 11$

and

- $x + y = 112$.

Find x and y.

$x =$ .....

$y =$ ..... [2]

14 **(a)** This is the term-to-term rule for a sequence.

Multiply by 2 and add 3

The first three terms in this sequence are 1, 5 and 13.

Write down the next term in this sequence.

..... [1]

(b) This is the term-to-term rule for a different sequence.

Square and subtract 5

The second and third terms in this sequence are -1 and -4.

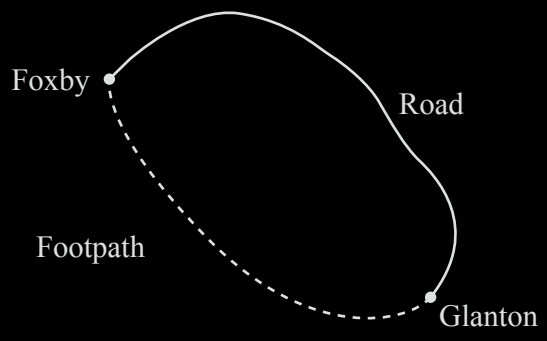
(i) Write down the fourth term in this sequence.

..... [1]

(ii) Write down the two possible values for the first term in this sequence.

..... or ..... [2]

15



NOT TO SCALE

Two villages, Foxby and Glanton, are joined by a footpath and a road.

(a) Abdul walks along the footpath from Foxby to Glanton.
He walks for 2 hours 14 minutes and arrives at Glanton at 15 10.

Calculate the time Abdul left Foxby.

..... [1]

(b) The distance, by road, between Foxby and Glanton is 15 km.
A bus travels along the road between Foxby and Glanton.
The bus journey takes 12 minutes.

Calculate the average speed of the bus in kilometres per hour.

..... km/h [2]

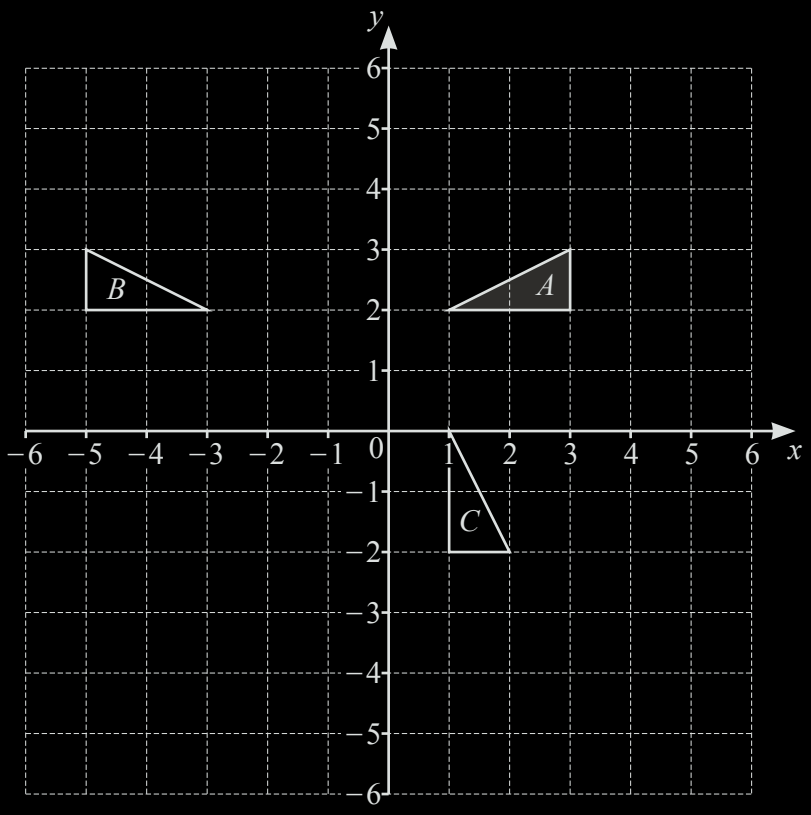
(c) The bearing of Glanton from Foxby is 128°.

Calculate the bearing of Foxby from Glanton.

..... [1]

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16



Triangles A, B and C are drawn on the grid.

(a) Describe fully the **single** transformation that maps triangle A onto triangle B.

.....

..... [2]

(b) Describe fully the **single** transformation that maps triangle A onto triangle C.

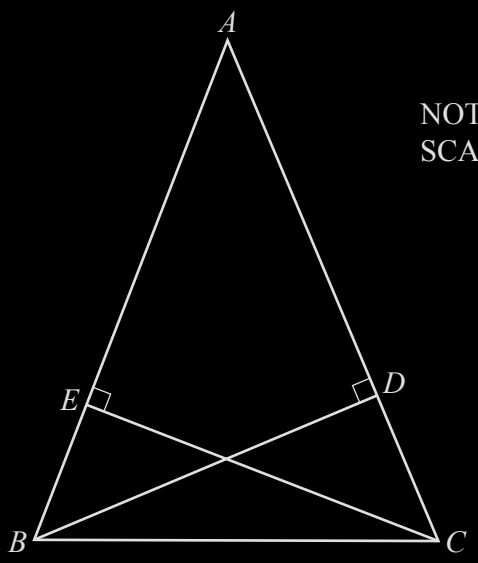
.....

..... [3]

(c) Triangle D is the image of triangle A after an enlargement, scale factor 2, with centre of enlargement (1, 2).

Draw triangle D. [2]

17



The diagram shows an isosceles triangle ABC where $AB = AC$.
D is a point on AC such that angle $ADB = 90°$.
E is a point on AB such that angle $AEC = 90°$.

Show that triangles ADB and AEC are congruent.
Give a reason for each statement you make.

.....

.....

.....

.....

..... [3]

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18 Solve the simultaneous equations.
Show your working.

$$x + 6y = 0$$
$$3x - 2y = 10$$

$x =$ .....

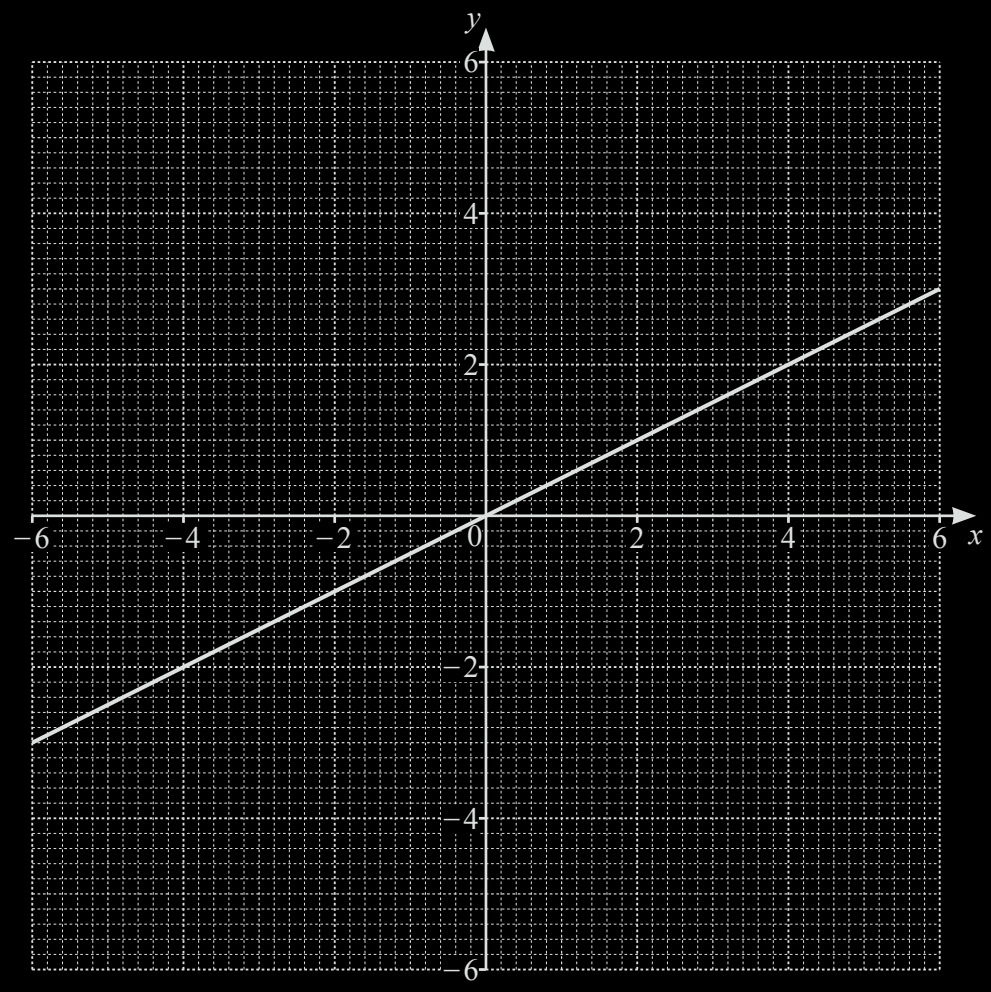
$y =$ ..... [3]

19 y is proportional to $(x-1)^2$.

Given that $y = 18$ when $x = 4$, find y when $x = 6$.

$y =$ ..... [2]

20



The line $2y = x$ is drawn on the grid.

(a) On the grid, draw the graph of

(i) $y = 2$, [1]

(ii) $y + x = 4$. [1]

(b) On the grid, shade and label the region **R**, defined by the following inequalities.

$x + y \leqslant 4$ $2y \geqslant x$ $y \leqslant 2$ $x \geqslant 0$ [2]

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21 Factorise.

(a) $3cx + 2bx - 6cy - 4by$

..... [2]

(b) $6x^2 + 7x - 10$

..... [2]

22 A car has a mass of 2400 kg, correct to the nearest hundred kilograms.
A caravan has a mass of 1460 kg, correct to the nearest ten kilograms.

Calculate the lower bound for the total mass of the car and caravan.

..... kg [2]

23 (a) $a = \dfrac{b^2 + c}{d}$

(i) Find a when $b = 4 \times 10^2$, $c = 6 \times 10^3$ and $d = 2 \times 10^2$.
Write your answer in standard form.

$a =$ ..... [3]

(ii) Rearrange the formula to make b the subject.

$b =$ ..... [3]

(b) $m \times 10^4 + m \times 10^2 = 36\,360$

Work out $m \times 10^4 - m \times 10^2$.

..... [2]

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24 **(a)** $\mathbf{M} = \begin{pmatrix} 5 & 1 \\ 2 & 3 \end{pmatrix}$ $\mathbf{N} = \begin{pmatrix} 4 & -2 \\ 3 & 0 \end{pmatrix}$

Find $\mathbf{M} - \mathbf{N}$.

$\begin{pmatrix} & \\ & \end{pmatrix}$ [1]

(b) $\mathbf{P} = \begin{pmatrix} 2 & 4 \\ c & -5 \end{pmatrix}$ $\mathbf{Q} = \begin{pmatrix} 3 & 2 \\ -2 & d \end{pmatrix}$ $\mathbf{PQ} = \begin{pmatrix} -2 & 0 \\ 19 & 11 \end{pmatrix}$

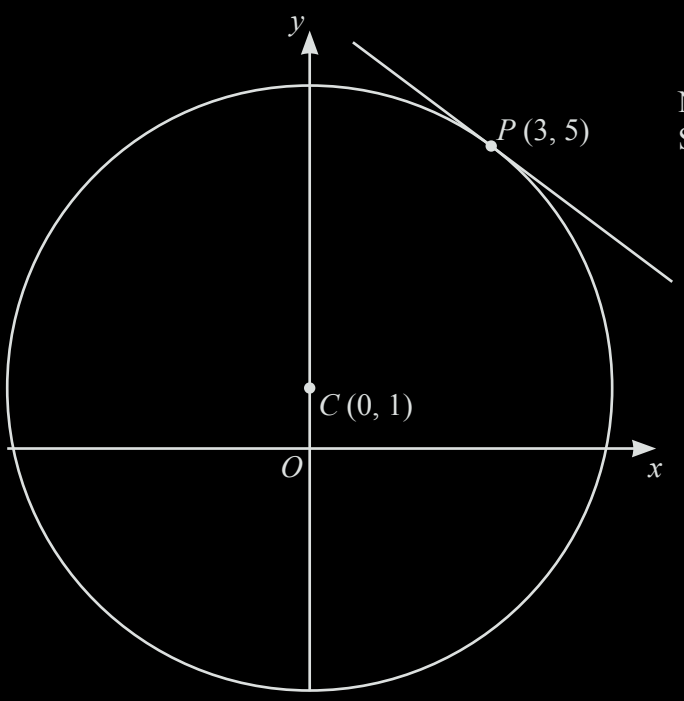
Find the value of c and the value of d.

$c =$ .....

$d =$ ..... [2]

Question 25 is printed on the next page.

25



The diagram shows a circle centre C (0, 1).
P (3, 5) is a point on the circumference of the circle.

Find the equation of the tangent at P.

..... [4]

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